Hongyi Xie （谢泓毅）

Research Fellow, Nanyang Technological University, Singapore

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Google Scholar: https://scholar.google.com/citations?user=7LgFC9wAAAAJ

Employment

Research Fellow *Singapore, Jul 2026 -now*

Asian School of the Environment, Nanyang Technological University

Engineer *China, Jul 2020 - Jul 2022*

Transport Planning and Research Institute, Ministry of Transport

Research Assistant *China, Jul 2015 - Sep 2015*

National Taiwan University

Education

Fudan University *China, Sep 2022 - Jun 2026*

Ph.D., with honors, Environmental Management, GPA: 3.26/4.0

Dissertation Topic: *Multi-Scale Effects and Synergistic Mechanisms of Carbon and Air Pollutant Emission Reductions from the Electric Passenger Vehicle Deployment in China*

National University of Singapore *Singapore, Aug 2024 - Aug 2025*

Joint Ph.D., Civil and Environmental Engineering

Beijing Normal University *China, Sep 2017 - Jun 2020*

M.S., with honors, Environmental Science, GPA: 88.35/100

Zhejiang University *China, Sep 2013 - Jun 2017*

B.S., Environmental Science, GPA: 3.45/4.0

Research Interest

- Energy-climate-environment nexus
- Co-benefits of cross-sector climate policies
- Constraints on renewable energy scale-up
- Integrated Assessment Models

Publications

H-index:11, total citations of 261 based on Google Scholar as 2026/08/21.

[#] denotes equal contribution; [*] denotes the corresponding author.

Peer reviewed:

Selected:

1. **Xie, H.**#, Wu, Y.#, Wang, Y.*, Guo, Y.*. (2026). Addressing water resource constraints for electrolytic hydrogen demand in China. ***Nature Sustainability***. https://doi.org/10.1038/s41893-026-01894-9

2. **Xie, H.**, Chen, B.*, Wang, C., Xie, W., Long, X., Zhang, D., Dai, M., Han, Z., Song, Y., Li, J., Wang, Y.*. (2025). Uncertain cross-sector climate actions could undermine air pollution reduction co-benefits in China's power and passenger car sectors. ***Engineering***, In press. https://doi.org/10.1016/j.eng.2025.09.006

3. **Xie, H.**, Chen, B., Dai, M., Han, Z., Bai, Y., Wang, Y.*. (2024). Upgrading passenger vehicle emission standard help to reduce China's air pollution risk from uncertainty in electrification. ***Environmental Science & Technology***, 58, 5325-5335. (Supplementary cover paper). https://doi.org/10.1021/acs.est.3c10078

4. **Xie, H.**, Li, H., Wang, S., Han, Z., Chen, B., Lv, H., Wang, Y.*. Spatial mismatches constrain high-value utilization of retired batteries for decarbonization in China. ***Environmental Science and Ecotechnology***. In press. https://doi.org/10.1016/j.ese.2026.100754

Others:

19. Wang, C., Xue, J., **Xie, H.**, Phillips, F., Liu, G., Zhang, F.***.** (2026). Tiered standards, differential effects: a quasi-experimental study of China's vehicle emission policies. ***Transportation Research Part D: Transport and Environment***, 158, 105454. https://doi.org/10.1016/j.trd.2026.105454

18. Wang, C., Meng, L., Zeng, X.*, **Xie, H.** (2026). Multi-dimensional drivers and their interactions on carbon sink capacity in tropical forests: A case study of Xishuangbanna. ***Environment, Development and Sustainability***. In press. https://doi.org/10.1007/s10668-026-07773-9

17. Li, H., Chen, B., **Xie, H.**, Yu, H., Dai, M., Xue, M., Long, X., Wang, J., Han, Z., Sun, M., Wang, J., Guo, Y., Wang, Y.*. (2026). Coal phaseout undermines co-benefits of fly ash as a clinker substitute. ***the Innovation,*** 7, 101496. https://doi.org/10.1016/j.xinn.2026.101496

16. Shui, B.*, Cai, Z., **Xie, H.**, Liu, F., Luo, X.*. (2026). Harnessing Policy Synergies to Decarbonize China's Light-Duty Passenger Vehicle Fleet. ***Journal of Cleaner Production***, 538, 147236. https://doi.org/10.1016/j.jclepro.2025.147236

15. Long, X., Chen, B., Dai, M., **Xie, H.**, Chen, P., Meng, J., Wang, Y.*. (2025). Ambitious climate targets can make the phaseout of India's coal-fired power plants cost-beneficial. ***Nature Communications***, 16, 11594. https://doi.org/10.1038/s41467-025-66580-4

14. **Xie, H.**#, Li, Y.#, Wang, C.*. (2025). Decarbonizing inland waterway transport: scenario analysis and mitigation strategy for Jiangxi Province, 2019–2050. ***ENGINEERING Environment***, 19(8), 113. https://doi.org/10.1007/s11783-025-2033-4

13. Zhang, Y., Dai, M.*, Chen, B., **Xie, H.**, Li, Z., Wang, Y.*. (2025). A Product-based Life-cycle Integration Platform for sustainable plastic bottles from agricultural biowaste. ***Chemical Engineering Journal***, 524, 169320. https://doi.org/10.1016/j.cej.2025.169320

12. Bai, Y., Li, H., Chen, B., **Xie, H.**, Wang, Y.*. (2025). Managing nitrogen metabolism of animal husbandry and aquaculture could mitigate nitrogen threat in the Main Cities of Yellow River Delta. ***Cleaner Environmental Systems***, 17,100273. https://doi.org/10.1016/j.cesys.2025.100273

11. Wang, C., Kong, Y., Lu, X., **Xie, H.**, Teng, Y., Zhan, J.*. (2024). Rethinking Regional High-Quality Development Pathways from a Carbon Emission Efficiency Perspective. ***Land***, 13(9), 1441. https://doi.org/10.3390/land13091441

10. Li, H., Bian, Y., Liu, M., Lin, J., Dai, M., **Xie, H.**, Yu, H., Chen, B., Xue, M., Li, Z., Yin, J., Xue, L.*. (2024). CO_2 mineralization and utilization by tailings sand in China for potential carbon sinks and spatial project layout. ***Resources, Conservation and Recycling***, 206, 107598. https://doi.org/10.1016/j.resconrec.2024.107598

9. Xie, W., Yang, X., Han, Z., Sun, M., Li, Y., **Xie, H.**, Yu, H., Chen, B., Fath, B.*, Wang, Y.*. (2024). Urban sector land use metabolism reveals inequalities across cities and inverse virtual land flows. ***Resources, Conservation and Recycling***, 202, 107394. https://doi.org/10.1016/j.resconrec.2023.107394

8. Fang, Y., **Xie, H.**, Chen, B., Han, Z., An, D., Cai, W., Zhang, W., Wang, Y.*. (2024). Achieving carbon neutrality in Shanghai's municipal wastewater treatment sector requires coordinated water conservation and technical improvement. ***Journal of Cleaner Production***, 443, 141134. https://doi.org/10.1016/j.jclepro.2024.141134

7. Han, Z., Xie, W., Yu, H., **Xie, H.**, Li, Y., Wang, Y.*. (2024). Rethinking industrial land-use in American rust cities towards sustainability based on a block-level model. ***Journal of Environmental Management***, 352, 120067. https://doi.org/10.1016/j.jenvman.2024.120067

6. Dai, M., Sun, M., Chen, B., **Xie, H.**, Zhang, D., Han, Z., Yang, L., Wang, Y.*. (2023). Advancing sustainability in China's pulp and paper industry requires coordinated raw material supply and waste paper management. ***Resources, Conservation and Recycling***, 198, 107162. https://doi.org/10.1016/j.resconrec.2023.107162

5. Gao, Y., Yi, Y.*, Chen, K., **Xie, H.** (2023). Simulation of suitable habitats for typical vegetation in the Yellow River Estuary based on complex hydrodynamic processes. ***Ecological Indicators***, 154, 110623. https://doi.org/10.1016/j.ecolind.2023.110623

4. Liang, Z., Deng, H., **Xie, H.**, Chen, B., Sun, M., Wang, Y.*. (2023). Rethinking the paper product carbon footprint accounting standard from a life-cycle perspective. ***Journal of Cleaner Production***, 393, 136352. https://doi.org/10.1016/j.jclepro.2023.136352

3. **Xie, H.**, Li,Y.*. (2022). An ecological water replenishment model of urban lake riparian plant restoration based on the groundwater – vegetation interactions. ***Ecological Engineering***, 176, 106510. https://doi.org/10.1016/j.ecoleng.2021.106510

2. **Xie, H.**, Yi, Y.*, Hou, C., Yang, Z. (2020). In situ experiment on groundwater control of the ecological zonation of salt marsh macrophytes in an estuarine area. ***Journal of Hydrology***, 585, 124844. https://doi.org/10.1016/j.jhydrol.2020.124844

1. Yi, Y.*, **Xie, H.**, Yang, Y., Zhou, Y., Yang, Z. (2020). Suitable habitat mathematical model of common reed (Phragmites australis) in shallow lakes with coupling cellular automaton and modified logistic function. ***Ecological Modelling***, 419, 108938. https://doi.org/10.1016/j.ecolmodel.2020.108938

In Chinese:

5. Wang, C., **Xie, H.**, Zhang, X., Zeng, X.*. (2025). Spatiotemporal variations of carbon source-sink matching in the Beijing-Tianjin-Hebei urban agglomeration. ***Chinese Journal of Ecology (生态学杂志)***, 44 (04), 1343-1354. (In Chinese). https://doi.org/10.13292/j.1000-4890.202504.041

4. Wang, S., Chen, B., Dai, M., **Xie, H.**, Zhang, D., Sun, M., Wang, Y.*. (2025). Carbon footprint accounting and impact factor analysis of China's bioeconomy. ***Environmental Engineering (环境工程)***,43(8), 233-243. https://doi.org/10.13205/j.hjgc.202508022

3. **Xie, H.**, Liu, J., Gao, Y., Li, Q., Gao, J., Yang, S., Zhu, G.*. (2022). Method and Application of Regional Highway Resources and Environment Carrying Capacity Evaluation. ***Highway (公路)***, 67 (09), 407-412.

2. Yi, Y.*, **Xie, H.**, Song, J., Yang, Z. (2021). Simulation of salt marsh vegetation community's suitable habitat in Yellow River Estuary I: theory. ***Journal of Hydraulic Engineering (水利学报)***, 52(03), 255-264. https://doi.org/10.13243/j.cnki.slxb.20200542

1. Yi, Y.*, **Xie, H.**, Song, J., Yang, Z. (2021). Simulation of salt marsh vegetation community's suitable habitat in Yellow River Estuary Ⅱ:application. ***Journal of Hydraulic Engineering (水利学报)***, 52 (04), 401-408. https://doi.org/10.13243/j.cnki.slxb.20200544

Working paper:

1. Xie, G., Bai, Y., Wang, Q., **Xie, H.***. Rethinking Urbanization under the Socio–Economic–Ecosystem Nexus: Insights from an Emergy Perspective in Shandong Province, China. (Major revision for ***Ecological Modelling***)

2. Fu, Y., Li, H., Chen, B., **Xie, H.**, Dai, M., Han, Z., Sun, L., Xue, M., Wang, Y.*. Spatial mismatch of biomass supply constrains biochar-based decarbonization in China's iron and steel industry. (Major revision for ***Resources, Conservation & Recycling***)

3. Sun, C., Han, Z., **Xie, H.**, Li, H, Chen, B., Zhao, Y., Wang, Y.*. Spatial Mismatch Embedded in the UK's Sectoral Land Use Efficiency. (Under review for ***Resources, Conservation & Recycling***)

4. Han, Z., Chen, B., Chen, Z., Shi, Z., Wang, S., Song, Y., **Xie, H.**, Li, H., Sun, C., Meng, J., Wu, L.*, Wang, Y.*. Location-sector pairing redefines sectoral flood exposure and network propagation across Chinese cities. (Under review for ***Engineering***)

5. Wang, S., Chen, B., Tang, Z., Pradhan, P., Warchold, A., Dai, M., Xie, H., Yan, J., Sun, C., Tu, Q.*, Wang, Y.*. Global non-food bioeconomy policies are fragmented, with governance and social sustainability gaps. (Submitted to ***Nature Sustainability***)

6. Wang, Z., Li, Y.*, **Xie, H.**, Zou, T., Yang, Y. Spatial heterogeneity of life cycle carbon emissions from hydrogen production in China. (Submitted to ***Energy***)

7. Li, H., Fu, Y., Dai, M., **Xie, H.**, Han, Z., Xue, M., Chen B.*, Song, X., Sun, C., Wang, S., Cao, D., Wang, Y.*. Supply-limited substitutes undermine cement decarbonization in China. In preparation.

8. Han, Z.#, Song, Y.#, **Xie, H.#**, Chen, B., Wang, Y.*. Local final demand drives China's sectoral land-use-induced biodiversity loss. In preparation.

9. Sun, C., Xie, W., Shui, B., Wang, S., **Xie, H.***. Quantifying the highways network slope photovoltaic power potential in China. In preparation.

Reports, books and standards:

1. United Nations Environment Programme (2024). Global Bioeconomy Assessment: Coordinated Efforts of Policy, Innovation, and Sustainability for a Greener Future. https://wedocs.unep.org/20.500.11822/45332

2. China Environment Publishing Group (2020). Theory and Practice of Risk Analysis and Emergency Capacity Planning for Oil Spills on Water. ISBN: 978-7-0000-0000-0. https://xueshu.baidu.com/ndscholar/browse/detail?paperid=1h2s0ge02e4d0rf0mk1r0xg0qm016243

3. Chinese Group Standard of China Institute of Navigation (2023). Guidelines for the construction planning of ship oil spill response capacity of inland water (T/CIN 025-2023). https://www.cinnet.cn/zh-hans/system/files/files/25.nei_he_chuan_bo_yi_you_ying_ji_neng_li_jian_she_gui_hua_bian_zhi_zhi_nan_.pdf

AWARDS

- Outstanding Graduate, Fudan University (2026)
- Joint PhD Training Program for State-constructed High-level Universities, China Scholarship Council (2024)
- Second-class Academic Scholarship, Fudan University (2024, 2025)
- Third Prize, Excellent Consulting Achievement Award in Port and Waterway Engineering, China Waterborne Transport Construction Association (2023)
- Outstanding Graduate, Beijing Normal University (2020)
- First-/Second-class Academic Scholarship, Beijing Normal University (2018, 2019)

CONFERENCES & PRESENTATIONS

2025 – 12th International Conference on Industrial Ecology, Singapore – Oral Presentation

2024 – Nature Conference on Air Pollution and Climate Change, Beijing – Poster Presentation

2023 – International Conference on Resource Sustainability, United Kingdom – Oral Presentation

2018 – 1st International Student Forum on Environmental Ecology, Beijing – Poster Presentation

TECHNICAL SKILLS

Analysis: Techno-economic assessment; Life Cycle Assessment; Prospective LCA (premise); Material Flow Analysis; Emission inventory; Spatial Analysis (ArcGIS)

Modeling: Integrated Assessment Model (GCAM); Multi-Agent modeling (AnyLogic); Hydrodynamic modeling (MIKE 21, Visual MODFLOW)

Coding: Matlab, R, Python

SERVICE

Journal referee:

Nature Cities; Environmental Science & Technology; Resources, Conservation and Recycling; Applied Energy; Journal of Cleaner Production; Journal of Industrial Ecology; Ecological Economics; Journal of Environmental Management; Scientific Data; Geography and Sustainability; Ecological Indicators; iScience; Journal of Hydrology: Regional Studies; Regional Studies in Marine Science; Journal of Sea Research; Heliyon; Energy, Ecology and Environment; Results in

Engineering; Bioresource Technology Reports; Environmental Development; Research in Transportation Business & Management; Unconventional Resources; Energy Nexus; Environmental Nexus; Atmospheric Environment: X